1st Order DEs: Modelling

Questions 1–46

Question 1. A bucket contains V litres of water at time t minutes. Water flows into the bucket at a constant rate. Write down a differential equation for:

a) $\frac{dV}{dt}$

b) A bucket contains V litres of water at time t minutes. The bucket leaks at a rate proportional to the volume of water present. Write down a differential equation for $\frac{dV}{dt}$.

c) A tank contains x litres of water at time t minutes. Water is pumped into the tank at a constant rate of 6 litres per minute. Water leaks out at a rate proportional to the amount of water in the tank. Write down a differential equation for $\frac{dx}{dt}$.

d) A cistern is fitted with a control valve. Let V be the volume of water (in litres) at time t minutes. The valve acts so that the rate of change of V is proportional to the difference between 200 litres and V. Write down a differential equation for $\frac{dV}{dt}$.

e) A metal object has temperature T°C at time t minutes. It is placed in a room at a constant temperature of 18°C. The object cools at a rate proportional to the difference between its temperature and the room temperature. Write down a differential equation for $\frac{dT}{dt}$.

f) A lake contains P tonnes of pollutant at time t months. Pollutant enters the lake at a rate of $0.5t$ tonnes per month. Natural processes remove pollutant at a rate proportional to the amount present. Write down a differential equation for $\frac{dP}{dt}$.

Question 2. A cart of mass 6 kg moves along a straight horizontal track. At time t seconds its speed is v m s^{-1}. A motor provides a forward driving force of magnitude $(18 - 3t)$ N. A resistive force of magnitude $4v$ N acts opposite to the motion. Assuming the cart continues moving forwards, form a differential equation for v in terms of t.

Question 3. A sled of mass 10 kg is pulled along level snow. At time t seconds its velocity is v m s^{-1}. The pulling force is constant and equal to 25 N. The resistive force from the snow is modelled as having magnitude $(5 + 0.2t)v$ N, acting opposite to the direction of motion. Form a differential equation for v in terms of t.

Question 4. A skydiver of mass 80 kg jumps from a stationary helicopter and falls vertically downwards. At time t seconds, their speed is v m s^{-1}. The air resistance is modelled as a force of magnitude $6v$ N, opposing the direction of motion. Assume acceleration due to gravity is g m s^{-2}. Form a differential equation relating v and t.

Question 5. A particle of mass 2 kg moves along a straight horizontal line. At time t seconds, its velocity is v m s^{-1}. The particle is acted on by a driving force of magnitude $6e^{-t}$ N in the direction of motion, and a resistive force of magnitude v^2 N acting opposite to the motion.

a) Form a differential equation for v in terms of t.

b) Explain why the differential equation cannot be solved using the integrating factor method.

Question 6. A particle of mass 5 kg moves along a straight horizontal line. At time t seconds, its velocity is v m s^{-1}. The particle is acted on by a forward driving force of magnitude 8 N, and a resistive force of magnitude $2v\,\frac{dv}{dt}$ N acting opposite to the direction of motion.

a) Form a differential equation for v in terms of t.

b) Explain why the differential equation cannot be solved using the integrating factor method.

It is given that the initial velocity of the particle is 0.5 m s^{-1}.

c) Solve the differential equation, writing v in terms of t.

Question 7. A tank contains 80 litres of water which has 12 kg of salt dissolved into it. Find the concentration of salt in kg l^{-1}:

a) in the tank;

b) in terms of S and V;

c) A tank contains 200 litres of water which has 5 kg of salt dissolved into it. Liquid flows out of the tank at 3 litres per minute. Write down the rate (in kg/min) at which the salt leaves.

Question 8. A tank contains 100 litres of solution with S kg of salt dissolved into it. The mixture is stirred. Liquid flows out at 4 litres per minute. Write down the rate at which salt leaves the tank in terms of S.

Question 9. Brine enters a tank at 5 litres per minute with concentration 0.3 kg per litre. The mixture leaves at 5 litres per minute. Let S kg be the mass of salt in the tank. The volume of the tank is 100 litres.

a) Find the rate at which salt enters the tank.

b) Write the rate at which salt leaves the tank in terms of S.

c) Hence form a differential equation for $\frac{dS}{dt}$.

Question 10. A tank contains 50 litres of solution and initially 2 kg of salt is in the solution. Brine enters at 2 litres per minute with concentration 0.4 kg per litre, and the mixture leaves at 2 litres per minute. Let S kg be the mass of salt after t minutes. Form a differential equation for $\frac{dS}{dt}$ and state an initial condition for S.

Question 11. A tank initially contains 100 litres of water. Water flows into the tank at a rate of 3 litres per minute and flows out of the tank at a rate of 5 litres per minute. Write down an equation for the volume, V litres, of the water at time t minutes.

Question 12. A tank initially contains 60 litres of water. Water flows into the tank at a rate of 7 litres per minute and flows out of the tank at a rate of 2 litres per minute. Write down an equation for the volume, V litres, of the water at time t minutes, and state a suitable domain.

Question 13. A tank initially contains V_0 litres of water. Water flows into the tank at a rate of p litres per minute and out of the tank at a rate of q litres per minute. Write down an equation for the volume, V litres, of the water at time t minutes.

Question 14. A tank contains 100 litres of solution with S kg of salt dissolved into it. Pure water enters the tank at a rate of 4 litres per minute and the mixture leaves the tank at a rate of 4 litres per minute. Write down a differential equation for $\frac{dS}{dt}$.

Question 15. A tank initially contains 100 litres of solution with S kg of salt mixed into it. Pure water enters the tank at a rate of 4 litres per minute and the mixture leaves the tank at a rate of 6 litres per minute.

a) Find an equation for V, the volume of salt solution, in the tank at time t minutes.

b) Write down the rate at which salt leaves the tank.

c) Hence form a differential equation for $\frac{dS}{dt}$.

Question 16. A tank initially contains 100 litres of solution with 6 kg of salt dissolved in it. Brine of concentration 0.2 kg per litre enters the tank at 4 litres per minute, and the mixture leaves the tank at 6 litres per minute. Let S kg be the mass of salt in the tank after t minutes. Form a differential equation for $\frac{dS}{dt}$.

Question 17. A tank initially contains 40 litres of solution with 5 kg of salt dissolved into it. Brine of concentration 0.5 kg per litre enters the tank at 3 litres per minute and flows out at 2 litres per minute. Let S kg be the mass of salt in the tank after t minutes.

a) Find an equation for V, the volume of solution in the tank at time t minutes.

b) Form a differential equation for $\frac{dS}{dt}$.

c) Explain briefly what happens to the concentration as $t \to \infty$.

Question 18. A student models the temperature T°C of a cup of tea after t minutes using $\frac{dT}{dt} = -0.12(T - 18)$.

a) Explain what the constant 18 represents.

b) Explain what the term $(T - 18)$ represents.

c) Explain why the negative sign is sensible.

d) Explain what the constant 0.12 represents.

Question 19. A population of fish, P, is measured in thousands. Time t is measured in years. A student models the population using $\frac{dP}{dt} = 0.4P - 2$.

a) Explain what the term $0.4P$ represents.

b) Explain what the term -2 represents.

c) Explain why the term -2 does not depend on P.

Question 20. The amount of charge stored in a capacitor at time t seconds is Q, measured in coulombs (C). The rate at which charge is added is called the charging rate and is given by $\frac{dQ}{dt} = 6 - \frac{Q}{4}$. A student models the charging process using:

a) Explain the term 6 represents.

b) Explain what the term $\frac{Q}{4}$ represents.

c) Explain why the sign of the $\frac{Q}{4}$ term makes sense physically.

d) Explain what happens to $\frac{dQ}{dt}$ as Q becomes large.

Question 21. A tank contains 100 litres of liquid, which is always well mixed. Let S kg be the mass of salt in the tank after t minutes. It is given that S satisfies $\frac{dS}{dt} = 0.8 - \frac{5S}{100}$.

a) What does the term 0.8 represent?

b) What does the term $\frac{5S}{100}$ represent?

c) State the outflow rate in litres per minute.

d) State the inflow concentration of salt in kg per litre.

Question 22. A gas tank contains a mixture of gases. Let x be the volume of oxygen (cm^3) after t hours. Suppose x satisfies $\frac{dx}{dt} = 3.5 - \frac{2x}{50+4t}$.

a) What does 3.5 represent?

b) Explain why the term $\frac{2x}{50+4t}$ represents oxygen leaving the tank.

c) Find the time at which the volume of oxygen is decreasing at a rate of 0.5 cm^3 h^{-1}. Give your answer in minutes.

Question 23. A well-mixed tank contains salt solution of constant volume 40 litres. Solution flows out of the tank at a rate of 3 litres per minute. Let S be the mass of salt in the tank, measured in kg, after t minutes. A student proposes the model $\frac{dS}{dt} = 0.6 - \frac{3S}{40}$.

a) State the units of $\frac{dS}{dt}$.

b) State the units of the constant term 0.6.

c) Show that the term $\frac{3S}{40}$ has the correct units.

Question 24. A tank contains a liquid of volume V litres at time t minutes. A chemical is dissolved in the liquid. Let x be the mass of the chemical in the tank, measured in grams. The volume of liquid in the tank is modelled by $V = 80 + 2t$. A student writes the differential equation $\frac{dx}{dt} = 1.5 - \frac{4x}{80+2t}$.

a) The student claims that the constant term 1.5 represents the chemical entering the tank. What must the units of 1.5 be?

b) Determine the units of the number 4.

c) Explain why the equation is dimensionally inconsistent if x were measured in kg instead of grams, unless the constant term were adjusted.

Question 25. A population of bacteria is being studied. Let P be the population at time t hours, measured in thousands of bacteria. A student proposes the model $\frac{dP}{dt} = k - \frac{P}{12+t}$.

a) State the units of $\frac{dP}{dt}$.

b) Hence state the units of k.

c) Explain why the number 12 must have units, and state its units.

d) The student later changes the time unit so that t is measured in minutes instead of hours. Write the model in terms of the new variable T, where T is time in minutes. (You may leave k in terms of its original units but the equation must be dimensionally consistent.)

Question 26. A tank initially contains 90 litres of water. Water drains from the tank at a constant rate of 3 litres per minute. Let V be the volume of water (in litres) after t minutes.

a) Write down an expression for V in terms of t.

b) State the domain of t for which the model is physically valid.

c) Explain why the model is not valid outside this domain.

Question 27. A tank initially contains 40 litres of water. Water flows into the tank at 6 litres per minute and flows out of the tank at 2 litres per minute. The tank has a maximum capacity of 120 litres. Let V be the volume (in litres) of liquid in the tank after t minutes.

a) Find $V(t)$.

b) Find the time when the tank begins to overflow.

c) Hence state the domain of t for which the model is valid.

d) Give one limitation of the model.

Question 28. A population of rabbits is modelled by $\frac{dP}{dt} = 0.12P$, where P is the population size after t years.

a) Solve the differential equation, given that $P(0) = 500$.

b) Use your model to predict the population after 20 years, giving your answer to three significant figures.

c) Explain why this prediction may not be realistic.

d) State one improvement that could be made to the model.

Question 29. A cup of coffee is modelled by $\frac{dT}{dt} = -0.18(T - 20)$, where T is the temperature of the coffee in °C and t is the time in minutes.

a) Explain why the model predicts that T will never be below 20°C.

b) Solve the differential equation, given that $T = 70$ when $t = 0$.

c) State two assumptions made in this model.

Question 30. A chemical is placed in a container and decays according to the model $\frac{dM}{dt} = -\frac{5M}{100-t}$, where M is the mass of chemical (in grams) at time t minutes.

a) State the domain of t for which the model is valid.

b) Explain why the model cannot be valid for all $t \geq 0$.

c) Solve the differential equation, given that $M(0) = 80$.

d) In the domain of t for which the model is valid, use your solution to describe what happens to M as $t \to 100$.

e) Suggest one reason why the model might be unrealistic.

Question 31. A tank contains 120 litres of water in which 10 kg of salt is dissolved. At time t minutes, brine containing 0.25 kg of salt per litre flows into the tank at a rate of 4 litres per minute. The mixture is kept uniform by stirring and flows out of the tank at a rate of 4 litres per minute. Let S kg be the mass of salt in the tank after t minutes.

a) Show that S satisfies the equation $\frac{dS}{dt} = 1 - \frac{S}{30}$.

b) Solve the differential equation.

c) Find the mass of salt in the tank after 20 minutes.

Question 32. A tank contains 50 litres of solution with 5.5 kg of salt dissolved in it. At time t minutes, brine containing 0.25 kg of salt per litre flows into the tank at a rate of 4 litres per minute. The mixture is stirred and flows out of the tank at a rate of 6 litres per minute. Let S kg be the mass of salt in the tank after t minutes.

a) Show that S satisfies the differential equation $\frac{dS}{dt} = 2 - \frac{3S}{25+t}$.

b) Solve the differential equation.

c) Find the concentration as $t \to \infty$.

d) State one limitation of the model.

Question 33. A pressure vessel initially contains 600 cm^3 of nitrogen. Gas leaks out of the vessel at a constant rate of 30 cm^3 per hour, and at the same time a gas mixture is pumped into the vessel at a rate of 70 cm^3 per hour. The gas being pumped in contains 8% carbon dioxide and 92% nitrogen. Assume the gases mix and uniformly inside the vessel. Let x cm^3 be the volume of carbon dioxide in the vessel.

a) Show that the situation can be modelled by the differential equation $\frac{dx}{dt} = 5.6 - \frac{3x}{600+40t}$.

b) Hence find the volume of carbon dioxide in the vessel after 5 hours, giving your answer to three significant figures.

c) Find the concentration (as a percentage) of carbon dioxide in the vessel as $t \to \infty$.

d) Describe one limitation of this model.

Question 34. Two different liquids, ethanol and water, are mixed together in a vessel. The contents are stirred continuously so that the mixture is always uniform. Initially, the vessel contains 15 litres of ethanol and 25 litres of water. From the time $t = 0$, ethanol is added to the vessel at a rate of 3 litres per second and water is added to the vessel at a rate of 1 litre per second. The well-mixed contents are removed at a rate of 4 litres per second. Hence the total volume in the vessel remains constant. Let x be the amount of ethanol in the vessel at time t seconds.

a) Show that x can be modelled by $\frac{dx}{dt} = 3 - \frac{x}{10}$.

b) Solve the differential equation to determine how long it will take for the ratio of ethanol to water in the vessel to be 2:1, according to the model. Give your answer to the nearest second.

c) It is observed experimentally that the solution takes a ratio of 2:1 of ethanol to water after 18 seconds. Use this information to comment on the validity of the model, giving a reason for your answer.

Question 35. A laboratory tank initially contains 5 litres of pure water. At time t minutes, a chemical solution is pumped into the tank at a constant rate of 100 ml per minute. At the same time, the well-mixed liquid leaves the tank through an outlet pipe at a constant rate of 50 ml per minute. The concentration of chemical in the incoming solution is not constant. It is modelled by $2 + 3e^{-0.08t}$ mg per litre. Let m be the mass of chemical (in mg) in the tank at time t minutes after pumping begins.

a) Find an expression for the volume of liquid in the tank, in litres, at time t.

b) Show that m satisfies a differential equation of the form $\frac{dm}{dt} + \frac{m}{100+t} = \frac{1}{10}(2 + 3e^{-0.08t})$.

Question 36. A metal rod is initially at temperature 0°C. Its temperature T at time t seconds is modelled by $\frac{dT}{dt} + T = 4e^t \cos t$. Find the temperature of the rod at time $t = \pi$, giving your answer in exact form.

Question 37. A company uses a subscription-based online service. Let N be the number of active subscribers at time t years.

a) The rate at which subscribers leave the service is proportional to the number of active subscribers. In one year, the number of subscribers lost is equal to 12% of the number at the start of the year. Assuming no new subscribers join, giving your answer to three significant figures, find the constant k such that $\frac{dN}{dt} = -kN$.

The company introduces a marketing campaign. The rate at which subscribers leave remains the same as in part a), it is also given new subscribers join at a rate of 30 per year and the company also gains 60 subscribers per year through a long-term partnership.

b) Write down a first-order differential equation for N in terms of t.

c) Initially, the company has 500 subscribers. Solve your differential equation from part b) to find N in terms of t.

d) Describe two simplifications that have been made in forming the model, that could affect its accuracy.

Question 38. A water treatment system is operating, and the mass of a chemical impurity, M grams, in the system is modelled by the differential equation $\left(\frac{4+2t}{\sqrt{t}}\right)\frac{dM}{dt} + \frac{4M}{\sqrt{t}} = 105(2+t)$, where t is the time in hours after the system starts operating. Initially, there are 4 grams of the impurity present.

a) Find, according to the model, the mass of impurity present after 6 hours.

b) Find, according to the model, the rate of change of the mass of impurity after 3 hours.

c) State one assumption of the model that may not be valid in practice.

Question 39. A pharmaceutical company is testing a drug in solution. Let x be the amount (in mg) of drug remaining in the solution after t minutes. The drug breaks down over time, and the rate of decrease of the drug in solution is proportional to $x^2 + 16$. Initially, the amount of the drug in the solution is 120 mg. After 50 minutes, this has reduced to 60 mg. Find the amount of drug remaining in the solution after 200 minutes.

Question 40. A particle of mass 3 kg moves along a straight horizontal line. At time t seconds, its velocity is v m s^{-1}. The particle is acted on by two forces: a driving force of magnitude $(6+2t)$ N acting in the direction of motion, and a resistive force of magnitude $3v$ N acting opposite to the motion.

a) By forming a differential equation, show that the motion of the particle satisfies $\frac{dv}{dt} + v = 2 + \frac{2}{3}t$.

Question 41. A small trolley of mass 0.4 kg moves along a straight horizontal track. At time t seconds, its speed is u m s^{-1}. Two horizontal forces act on the trolley: a driving force P N acting in the direction of motion, and a resistive force Q N opposing the direction of motion. In an initial model, P is assumed to be constant with a constant of proportionality k, and Q is assumed to be proportional to speed u with a constant of proportionality c, where $k, c > 0$.

a) Show that the situation can be modelled by $\frac{du}{dt} = \frac{5}{2}(k - cu)$.

b) Solve this differential equation to obtain u in terms of t, k, and c, given that the trolley starts from rest.

You are now given that $k = 1.6$ and $c = 0.8$.

c) Find the acceleration of the trolley when $t = 2$ seconds.

In a refined model, the driving force has constant magnitude 1.6 N, while the resistive force remains as it was in the original model.

d) Write down the differential equation satisfied by u in Model B.

e) By considering the sign of $\frac{du}{dt}$, explain why Model B predicts that u tends to a limiting value as $t \to \infty$, and find this limiting value.

Question 42. A particle of mass m kg is projected vertically downwards with initial speed u m s^{-1}. At time t seconds its velocity is v m s^{-1}, where v is taken as positive downwards. The acceleration due to gravity is g m s^{-2}.

a) In Model A, the resistive force is assumed to have magnitude kmv N, acting upwards, where k is a positive constant. Show that v satisfies $\frac{dv}{dt} + kv = g$.

b) Solve the differential equation to obtain v in terms of t.

c) Show that the particle approaches a terminal velocity, and state its value.

In Model B, the resistive force is instead assumed to have magnitude $km\sqrt{v}$ N, acting upwards.

d) Write down the differential equation satisfied by v in Model B.

e) By considering the sign of $\frac{dv}{dt}$, explain why Model B predicts that v tends to a limiting value as $t \to \infty$, and find this limiting value.

Question 43. A train carriage moves along a straight horizontal track. The total mass of the carriage is 17 tonnes. At time t seconds its velocity is v m s^{-1}, where $0 \leq t \leq 10$. A resistive force acts at all times, modelled as a force of magnitude $2.12v$ N, acting opposite to the direction of motion. The carriage starts from rest at $t = 0$. For the first part of the motion, the carriage is pulled forward by an engine. Later, the engine stops pulling, and the carriage is slowed down by an additional braking force. At $t = 10$, the carriage comes to rest.

a) State the initial magnitude of the resistive force.

From $t = 6$ to $t = 10$, the engine is no longer pulling. Instead, the buffer exerts a constant force k N opposite to the motion.

b) Form a differential equation satisfied by v for $6 \leq t \leq 10$.

c) Solve your differential equation from part b) to find v in terms of t.

d) Hence find the speed of the carriage at $t = 6$.

e) Find the time (in minutes) predicted by this model for the speed to reach half its limiting value.

During the interval $0 \leq t \leq 6$, the pulling force from the engine has magnitude ct N, where c is a positive constant.

f) Show that $\frac{dv}{dt} + 0.12v = \frac{c}{m}t$.

g) Solve this differential equation to obtain v in terms of t for $0 \leq t \leq 6$.

h) Use your answers to show that $\frac{c}{m} = \frac{27e^{0.48} - 27}{125e^{0.72} - 35}$.

i) Find the total distance travelled by the carriage between $t = 0$ and $t = 10$. Give your answer to the nearest metre.

Question 44. A manufacturer tests the speed controller of an electric turntable. Let v be the angular speed (in suitable units) at time t hours. Initially the turntable is stationary, and in the long-term the speed approaches a constant value V.

a) A basic model assumes the rate of increase of speed is proportional to the difference between the limiting speed and the current speed. Write down a differential equation for the model.

b) Verify that $v = V(1 - e^{-kt})$, where $k > 0$, satisfies the differential equation, the initial condition, and the long-term condition.

c) A different model suggests that v satisfies the linear differential equation $\frac{dv}{dt} + \frac{2t}{1+t^2}v = \frac{2Vt}{1+t^2} + F(t)$, where $F(t)$ is a function of t. Find an integrating factor for this equation, simplifying it as far as possible.

d) Suppose $F(t) = 0$. Solve the differential equation and show that $v = \frac{Vt^2}{1+t^2}$.

Question 45. A brief spike in the output signal S (in suitable units) from a sensor is to be modelled. The spike begins at $t = 0$ seconds, and initially the output is zero. After 2 seconds, the output is measured as $S = 8$. When the spike is occurring, S is modelled by the differential equation $\frac{dS}{dt} = S - \frac{3t^2}{\sqrt{1+t^2}}$. The spike is assumed to last until the output is again zero.

a) Solve the differential equation to obtain an expression for S in terms of t.

b) Find the length of time that the spike lasts according to the model. Give your answer in exact form.

c) Find, in exact form, the rate of increase, $\frac{dS}{dt}$, at the end of the spike.

Question 46. A particle of mass 2 kg moves along a straight line with velocity v m s^{-1} at time t seconds. When $t = 0$, the velocity is -2 m s^{-1}. For $0 \leq t \leq 2$, the resultant force on the particle is $F = \frac{8}{t^2+2t+2} - \frac{2v}{(t+1)(t^2+2t+2)}$ N.

a) Form a differential equation for this model.

b) Find the first time t for which the particle is stationary.